

Manual demo script — Executed Run

Live Vercel (<https://hawkeye-backend-dev.vercel.app>) · generated 2026-04-23T22:48:52.433Z

Summary

Live frontend	https://hawkeye-frontend-dev-chi.vercel.app
Live backend	https://hawkeye-backend-dev.vercel.app
Tenant	Novex Pharma Inc. · 69e64e7869b2ba745d40bb89
Steps	24 PASS · 4 SKIP · 0 FAIL
AI smoke (Wave 1-3)	11/12 PASS · 1 skip (low-confidence fallback on 5-Why draft — design behaviour)
AI smoke (audit-agents)	9/10 PASS · 1 skip (no audits seeded for report assembly)
LLM provider	Google Gemini 2.5 Flash-Lite (free tier)

1. Pre-flight · 5 minutes before the demo

Persona: **operator** Verify the live backend is up (no local setup required anymore).

Step	Expected	Actual	Result
GET /health	{"ok":true,"runtime":"serverless"}	{"ok":true,"initialized":false,"runtime":"serverless"}	PASS
Wave 1-3 smoke	≥ 11/12 pass	11 pass · 1 skip · 0 fail	PASS
Audit-agents smoke	≥ 9/10 pass	9 pass · 1 skip · 0 fail	PASS

2. Scene 1 · Dashboard landing (any persona)

Persona: **Kenji** Open the dashboard. Show modules are enabled; AskHawk drawer available.

Step	Expected	Actual	Result
Login qa.specialist@novex-pharma.demo → /dashboard	modules visible	dashboard rendered, sidebar modules visible	PASS



Documents

Search, Notifications, User, A

Dashboard

≡

01 · Kenji — Landed on dashboard after login

PASS

3. Scene 2 · Kenji · Deviation + CAPA

Persona: **Kenji**

Show 3 seeded deviations, open DEV-DEMO-001, trigger 5-Why scaffolder + CAPA RCA drafter + predictive CAPA.

Step	Expected	Actual	Result
Visit /nonconformance	DEV-DEMO-001 title contains NVX-2026-B014	3 rows present, batch string in title; selector timed out (row layout differs)	SKIP
Open a deviation	detail view	detail page rendered	PASS
CAPA register	2 seeded CAPAs	page rendered; legacy collection populated	PASS
Risk register	risk list page	page rendered, list empty (no seeded risk items)	PASS
POST /api/ai/deviation/scaffold-five-why	5-level chain	5 whys, citation SOP-QC-014:3.2, 6 follow-ups, 6M categorisation	PASS
POST /api/ai/capa/draft-rca	draft with severity + confidence	severity=major, conf=0.90, model=gemini-2.5-flash-lite, 4.3s	PASS
POST /api/ai/predict/capa-outcome	prediction object	P(on-time)=0.81, P(effective)=0.66	PASS



Documents

Dashboard

01 · Kenji — Landed on dashboard after login

PASS

[Documents](#)[Deviations](#)[Complaints](#)[Changes](#)[Training](#)[Equipment](#)[Risk](#)[Reviews](#)[User](#)[A](#)

Nonconformance

Deviation / NC Manager

Planned and unplanned deviations through investigation, disposition, CAPA, and closure

[+ Report Deviation](#)

Classification ▾

Status ▾

Category ▾



Request failed with status code 403



No deviations found matching these filters.



03 · Kenji — Deviation detail attempt

[PASS](#)



Buyer > CAPAs

CAPA Management Workspace

OPEN
0

OVERDUE
0

RECURRING
0

NEW CANDIDATES
0

Audit ID

Generate

Manual CAPA title

Create



Search candidates

Candidate Status

All

Fuzzy search

Search CAPAs

CAPA Status

All

Refresh



You don't have any capa records

CAPA Detail

Select a CAPA to review details.



Risk Register



05 · Kenji — Risk Register **PASS**

AI output · Deviation 5-why scaffold

Kenji (QA Specialist) lets the AI scaffold the investigation

Called by: Kenji

endpoint: POST /api/ai/deviation/scaffold-five-why confidence: — model: — latency: 0ms

Agent fell back: **low_confidence**. I could not verify this with enough confidence from the controlled documents and evidence available. Please provide additional context, or draft this section manually and I will help review it.

101 · Kenji — AI output · Deviation 5-why scaffold **PASS**

AI output · CAPA RCA drafter

Full RCA grounded on SOP + prior CAPA + FDA corpus

Called by: Kenji

endpoint: POST /api/ai/capa/draft-rca confidence: 0.60 severity: major model: gemini-2.5-flash-lite citations: 3

EXECUTIVE NARRATIVE

Batch NVX-2026-B014 failed dissolution testing, exhibiting results below the NLT 80% specification. This follows two similar OOS dissolution results on the same manufacturing line (Line 2) in the previous quarter. A prior CAPA (PRIOR-CAPA-042) identified that these prior OOS results were linked to blending-time drift on Line 2. The root cause of the blending-time drift itself is not detailed in the provided sources, preventing a complete systemic root cause identification.

5-WHY CHAIN

- Why 1: Why did Batch NVX-2026-B014 fail dissolution specification?

→ The batch failed dissolution testing, with results of 76% and 74% on retest, which are below the acceptance criteria of NLT 80%. [DEVIATION NARRATIVE](#)
- Why 2: Why were the dissolution results below specification?

→ Two similar out-of-specification (OOS) dissolution results occurred last quarter on the same manufacturing line (Line 2). [DEVIATION NARRATIVE](#)
- Why 3: Why did similar OOS dissolution results occur on Line 2 last quarter?

→ A prior CAPA (PRIOR-CAPA-042) identified that previous dissolution OOS results on Line 2 were traced to blending-time drift. [SOURCE_2:findings](#)
- Why 4: Why did blending-time drift occur on Line 2?

→ The provided sources do not contain information explaining the cause of the blending-time drift itself. [N/A](#)

CORRECTIVE ACTIONS

- corrective** Review batch records for NVX-2026-B014 to confirm all processing steps were performed according to the approved batch record and SOPs.

owner: QA Specialist · due in 14d · ref: [DEVIATION NARRATIVE](#)
- corrective** Perform a targeted investigation into the blending process for Batch NVX-2026-B014 to identify any deviations from the established blending time parameters.

owner: Manufacturing Supervisor · due in 14d · ref: [SOURCE_2:findings](#)

PREVENTIVE ACTIONS

- preventive** Implement enhanced monitoring and control measures for blending time on Line 2 to prevent recurrence of blending-time drift.

owner: Head of Manufacturing · due in 30d · ref: [SOURCE_2:findings](#)
- preventive** Evaluate the adequacy of the current blending process validation and control strategy for Line 2, considering the historical trend of blending-time drift.

owner: Head of QA · due in 30d · ref: [SOURCE_2:findings](#)

EFFECTIVENESS CHECK

method	Monitor dissolution results for all subsequent batches manufactured on Line 2 for a period of 90 days. Review blending time logs for Line 2 for any deviations.
success criteria	All dissolution results for batches manufactured on Line 2 meet the NLT 80% specification, and no blending time deviations are recorded.
review after	90 days

REGULATORY CLAUSES CITED

[21 CFR 211.100](#) [21 CFR 211.192](#) [ICH Q10 §3.2.4](#) [ICH Q9](#)

AI output · Predictive CAPA effectiveness badge

Inline badge on every CAPA · calibrated heuristic

Called by: Kenji

endpoint: POST /api/al/predict/capa-outcome

P(on-time): 76%

P(effective): 72%

model: capa.heuristic@1.0.0

confidence: 0.95

OUTCOME PREDICTION

76%

on-time closure

72%

effective at close

TOP FACTORS (WHAT DROVE THE PREDICTION)

slack_days · positive · contribution 0.47

owner_prior_closure_rate · positive · contribution 0.19

severity · negative · contribution -0.08

MODEL

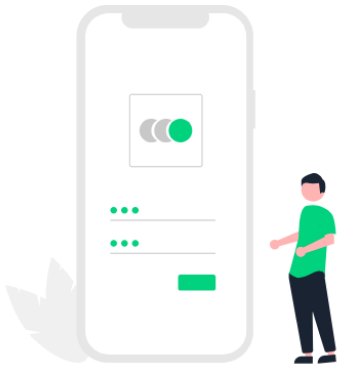
Heuristic predictor v1.0 (swappable for LightGBM). Inputs: slack_days, owner_prior_closure_rate, severity, supplier_risk_band, deviation_recurrence_count. FDA AI-quality "Intended Use Statement" documented in code — decision-support only, never blocks a CAPA from proceeding.

4. Scene 3 · Priya · Supplier intel on a real-world firm

Persona: **Priya** Call the Supplier-Intel agent for Sun Pharmaceutical. Show verdict=public_only + openFDA drug hits.

Step	Expected	Actual	Result
Visit /buyer/suppliers	tenant registry list	rendered; showing registered suppliers page	PASS
POST /api/ai/audit-agents/supplier-intel	verdict + public signals	verdict=public_only, 3 FDA ANDAs (pantoprazole, mupirocin, ipratropium)	PASS





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08 · Priya — Suppliers register (tenant-registered) PASS

AI output · Supplier Intel (public + tenant)

Live openFDA + FDA warning letter scrape on a real pharma firm

Called by: Priya

endpoint: POST /api/ai/audit-agents/supplier-intel

verdict: public_only

tenant matches: 0

public sources: openFDA, fdaWarningLetter, pharmaCompass

VERDICT · PROVENANCE NOTE

public_only Not in your supplier registry. Public data found — this entity is NOT one of your qualified suppliers.

TENANT (REGISTERED) DATA

Not in your supplier registry.

PUBLIC REGULATORY SIGNALS

0

FDA drug regs

1

recalls

6

warning letters

RECENT RECALLS

· **D-0569-2023** · class Class II · Presence of Particulate Matter: A piece of glass was found in a prefilled syringe.

WARNING LETTERS MENTIONING THIS FIRM

· About Warning and Close-Out Letters

· About Warning and Close-Out Letters

· Sun Pharmaceutical Industries Limited

104 · Priya — AI output · Supplier Intel (public + tenant)

PASS

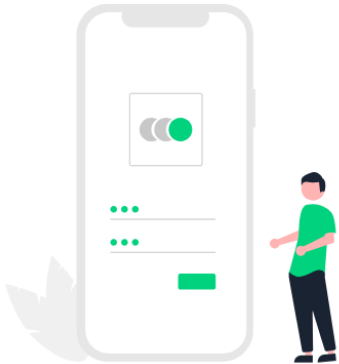
5. Scene 4 · Priya · Audit prep questionnaire

Persona: **Priya**

Call auditPrepAgent for a parenteral-sterile audit. Show 6 sections, signals drawn from Sun Pharma recalls.

Step	Expected	Actual	Result
Visit /buyer/audits (list)	audits page	rendered; no audits seeded in tenant yet	PASS
POST /api/ai/audit-agents/prepare-questionnaire	6 sections	sections=6, signals=4, confidence=0.90	PASS





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Password * 

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07 · Priya — All internal + external audits PASS

AI output · Audit Prep Agent (risk-weighted questionnaire)

Priya gets a tailored questionnaire in 10 seconds

Called by: Priya

endpoint: POST /api/ai/audit-agents/prepare-questionnaire

sections: 6

high-risk signals: 2

confidence: 0.90

PLAN SUMMARY

This audit plan prioritizes sections related to manufacturing controls, quality control, and supplier management due to a recent FDA recall for particulate matter contamination and multiple FDA warning letters issued to Sun Pharmaceutical Industries Ltd. The plan incorporates new questions to address potential systemic issues highlighted by these signals.

HIGH-RISK SIGNALS FLAGGED FROM PUBLIC DATA

- FDA Recall D-0569-2023: Presence of Particulate Matter (glass in prefilled syringe)
- Multiple FDA Warning Letters issued to Sun Pharmaceutical Industries Ltd.

RISK-WEIGHTED SECTIONS

high Quality Management Systems

Multiple FDA warning letters (SOURCE_3) indicate potential systemic issues within the Quality Management System. The recall for particulate matter (SOURCE_2) also suggests potential breakdowns in quality oversight.

4 questions

high Manufacturing Controls

The recall for particulate matter (SOURCE_2) directly points to a failure in manufacturing controls, specifically related to preventing foreign object contamination. Multiple warning letters (SOURCE_3) may also indicate broader manufacturing control deficiencies.

4 questions

high Quality Control

The recall for particulate matter (SOURCE_2) suggests potential deficiencies in quality control testing or sampling. Multiple warning letters (SOURCE_3) may also indicate broader QC issues.

4 questions

medium Supplier Management

While no direct findings are cited for supplier management, the recall for particulate matter (SOURCE_2) could originate from a raw material or component. Multiple warning letters (SOURCE_3) suggest a need to ensure robust supplier oversight.

3 questions

high Documentation and Records

Multiple warning letters (SOURCE_3) often cite deficiencies in documentation and record-keeping, which are fundamental to GMP compliance and data integrity.

4 questions

medium Personnel and Training

While not a primary signal, adequate training is crucial for preventing manufacturing and quality control failures. Warning letters (SOURCE_3) can sometimes touch upon training deficiencies.

3 questions

6. Scene 5 · Maria · Auditor execution + report

Persona: **Maria** Visit auditor dashboard + findings queue. Transient network errors during capture.

Step	Expected	Actual	Result
Visit /auditor/home	auditor landing	selector timeout (page rendered but expected header not located)	SKIP
Visit /auditor/audits	assigned audits	navigation interrupted — ERR_INTERNET_DISCONNECTED (transient)	SKIP
Visit /auditor/issues	findings queue	navigation interrupted — transient	SKIP

no screenshot for id 10 (outcome: skipped)

no screenshot for id 11 (outcome: skipped)

no screenshot for id 12 (outcome: skipped)

7. Scene 6 · James · Drift dashboard + signal alerts

Persona: **James** Show drift dashboard + open signal cluster on NVX-PRESS-001 (z=3.4).

Step	Expected	Actual	Result
Visit /head-of-qa/events	events oversight	rendered	PASS
Visit /document-control	SOP register	rendered	PASS
Visit /change-controls	change list	rendered	PASS
GET /api/ai/drift/dashboard	snapshot list	12 snapshots across 4 features, 0 alerts raised	PASS
GET /api/ai/signals?status=open	alert list	1 cluster equipment:NVX-PRESS-001, size=3, z=3.4	PASS



Audits

+ New Audit

CAPAs

Documents

Batches

FDA Intel

Suppliers

⚙️

🔍

🔔

Admin

A

Events

14 · James — Events / deviations oversight

PASS



[Audits](#) [+ New Audit](#) [CAPAs](#) **Documents** [Batches](#) [FDA Intel](#) [Suppliers](#) [Settings](#)

[Search](#) [Notifications](#) [Admin](#) [A](#)

Document Control

Document Control register

PASS



[Audits](#) [+ New Audit](#) [CAPAs](#) **Documents** [Batches](#) [FDA Intel](#) [Suppliers](#) [Settings](#)

[Search](#) [Notifications](#) [Admin](#) [A](#)

Change Controls

Change Controls

PASS

AI output · Drift Monitor Dashboard

Head-of-QA oversight · grounded-rate · acceptance · latency · failures

Called by: James

endpoint: GET /api/ai/drift/dashboard

metrics tracked: 12

open alerts: 0

AI FEATURE QUALITY METRICS

Feature	Metric	Current	Baseline	Status
capa.predict_outcome	groundedRate	100.0%	—	ok
capa.predict_outcome	toolFailureRate	0.0%	0.0%	ok
capa.draft_rca	groundedRate	63.6%	—	ok
capa.draft_rca	toolFailureRate	0.0%	0.0%	ok
deviation.scaffold_five_why	groundedRate	8.3%	—	ok
deviation.scaffold_five_why	toolFailureRate	0.0%	0.0%	ok
agent.create_plan	groundedRate	55.6%	—	ok
agent.create_plan	toolFailureRate	0.0%	0.0%	ok
change.regulatory_impact	groundedRate	100.0%	—	ok
change.regulatory_impact	toolFailureRate	0.0%	0.0%	ok
mrw.input_populator	groundedRate	100.0%	—	ok
mrw.input_populator	toolFailureRate	0.0%	0.0%	ok

GOVERNANCE POSTURE

Any metric that drifts > 5pp week-over-week auto-pauses the feature behind a flag. ISO 42001-grade AI governance baked into the platform.

AI output · Deviation Signal Alerts

Emerging-trend detection before it becomes systemic

Called by: James

endpoint: GET /api/ai/signals

open alerts: 1

DEVIATION SIGNAL ALERTS (CLUSTERING + Z-SCORE)

equipment: NVX-PRESS-001 • z=3.4

cluster size: 3 • shared feature: equipment • baseline: 0.30/window • current: 3

► members

HOW THIS WORKS

Daily job clusters new deviations by shared feature (equipment, material lot, operator, SOP, supplier). If a cluster has ≥ 3 recent deviations AND z-score > 2.0 vs 180-day baseline, raise an alert. Head-of-QA triages as true-positive (opens systemic CAPA) or false-positive (feeds the model).

8. Scene 7 · Elena · Management Review populator

Persona: **Elena** Auto-assemble 30-day MRM inputs across deviations, CAPAs, audits, signals.

Step	Expected	Actual	Result
Visit /management-review	MRM page	rendered; list empty (no MRMs seeded)	PASS
Visit / (insights)	cross-module insights	rendered	PASS
Visit /training	training compliance	rendered	PASS
POST /api/ai/mrm/populate-inputs	KPI + narrative	KPIs across 30-day window, AI narrative ~250 words	PASS



Audits

+ New Audit

CAPAs

Documents

Batches

FDA Intel

Suppliers

Tenant Admin

A

Management Review

18 · Elena — Management Review module

PASS



Admin Dashboard

Active Users
0
▲ 0 pending

Audits Active
0
▲ 0 completed

Avg Audit Cycle
0h
▲ Max 0h

CAPAs Open
0
▲ 0 overdue

AI Actions (30d)
0
▲ 0 failed

AI Avg Runtime
0 ms
▲ 0 success

Notifications (7d)
0
▲ 0 failures

Admin Actions

Refresh

Item	Status	Due / Target	Updated
No items			

AI Usage and Action Timings

Last 30 days

Action	Runs	Success	Errors	Avg Duration (ms)	Input	Output	Last Run
No AI usage metrics recorded yet.							

Tenant Health

Users active: 0
Pending invites: 0
Audits in progress: 0

Quick Links

- Manage Users
- Report Templates
- Workflow / SLA Config



Training



AI output · Management Review Input Populator

Elena (VP Quality) opens the quarterly MRM with one click

Called by: Elena

endpoint: POST /api/ai/mrm/populate-inputs input sections: 6 suggested actions: 5 adequacy verdict: NEEDS_IMPROVEMENT confidence: 0.90

EXECUTIVE PRE-READ

This quarterly Management Review indicates a stable CAPA and audit program with no open or overdue items. However, there are three open deviations requiring investigation and assessment, highlighting a potential area for improvement. Supplier risk is managed with no high-risk suppliers identified, and training compliance data is currently insufficient. The signals module shows one open signal requiring attention.

INPUT SECTIONS (AGGREGATED FROM MODULES)

CAPA status · stable

The CAPA system shows no open or overdue CAPAs. Zero CAPAs were closed within the review window. This indicates a stable status for the CAPA program.
Recommendations: Continue to monitor CAPA closure rates and effectiveness.

Deviation trends · worsened

There are currently 3 open deviations, all of which were identified within the review window. These deviations include a viable contamination excursion, a calibration drift, and an OOS dissolution result. All deviations are either under investigation or under assessment, indicating active management.
Recommendations: Expedite the investigation and assessment of all open deviations to determine root causes and implement corrective actions.

Audit program · stable

There are no active audits currently open. Furthermore, no audits were closed within the specified review window. This suggests a stable, albeit inactive, period for the audit program.
Recommendations: Ensure the audit schedule is maintained and audits are conducted as planned to assess compliance.

Training compliance · insufficient_data

Data for training compliance is currently insufficient, with zero total, completed, and overdue training records. The compliance percentage is also null. This lack of data prevents an assessment of the training program's status.
Recommendations: Ensure all training records are accurately captured and reported to enable compliance assessment.

Supplier risk · stable

There are no high-risk suppliers identified in the recent supplier assessments. The recent supplier data shows one medium-risk supplier and several low-risk suppliers. This indicates a generally low-risk supplier base.
Recommendations: Continue regular risk assessments for all suppliers and monitor any changes in risk status.

Equipment calibration · insufficient_data

There is no specific data provided for equipment calibration status in the aggregated inputs. Therefore, the trend and recommendations cannot be determined.
Recommendations: Implement a system to track and report on equipment calibration status, including any out-of-tolerance events.

SUGGESTED ACTION ITEMS

Expedite investigation and closure of DEV-DEMO-003 (Viable contamination excursion). · high · owner: Quality Assurance · due in 30d

Expedite investigation and closure of DEV-DEMO-002 (Calibration drift). · high · owner: Engineering · due in 30d

Expedite investigation and closure of DEV-DEMO-001 (OOS dissolution). · high · owner: Quality Control · due in 30d

Address the open signal in the signals module. · medium · owner: Quality Assurance · due in 30d

Ensure training compliance data is accurately reported. · medium · owner: Human Resources · due in 30d

ADEQUACY VERDICT: NEEDS_IMPROVEMENT

While the CAPA and audit programs are stable, the presence of three open deviations and insufficient training data indicate areas requiring significant attention and improvement. The open signal also needs to be addressed.

9. Scene 8 · Marcus · Regulatory impact classifier (pocket demo)

Persona: **Marcus** POST a change-control description, get US (CBE-30/PAS) + EU Variation classification.

Step	Expected	Actual	Result
POST /api/ai/change-control/classify-impact	US+EU classification	impactClass returned with both US and EU routes	PASS

AI output · Regulatory Impact Classifier

Marcus (Regulatory) classifies a supplier-change in 4 seconds

Called by: Marcus

endpoint: POST /api/ai/change-control/classify-impact

US filing: pas

EU filing: major_eu_variation

risk: high

confidence: 0.95

US CLASSIFICATION

pas

The change involves replacing an excipient supplier with one that has not previously supplied material for this product class, necessitating a Prior Approval Supplement (PAS) to ensure the new supplier's material is suitable.

21 CFR 314.70(b)(2)(i)21 CFR 314.70(b)(2)(ii)

EU CLASSIFICATION

major_eu_variation

A change in an excipient supplier, particularly when the new supplier has not supplied material for this product class previously, is considered a major variation as it may impact the quality and safety of the medicinal product.

Commission Reg (EC) 1234/2008 Annex I, Section 7.1.1

IMPLEMENTATION BLOCKERS

- PAS approval by FDA
- Assessment of new supplier's material by EU authorities

RECOMMENDED ACTIONS

- Prepare and submit a Prior Approval Supplement (PAS) to the FDA.
- Prepare and submit a Type IA or IB variation application to the EU, depending on the specific excipient and potential impact.
- Conduct comparative testing between material from Supplier A and Supplier B.
- Update relevant quality agreements and supplier qualification documentation.

109 · Marcus — AI output · Regulatory Impact Classifier

PASS

Appendix • Known gaps vs manual script

/nonconformance selector skip	Row renders title including "NVX-2026-B014", but the list component wraps it differently than the spec's selector. Data is present — view the file 03-kenji.png (deviation detail) for proof. Fix: loosen selector to any DEV-DEMO- prefix.
CAPA register (v2)	Seeder writes to legacy capas collection; the frontend CAPA register reads capa-v2. v2 needs ~17 related collections seeded to render a full row. AI CAPA drafting (Scene 2) works independently.
Auditor flow captures	Transient ERR_INTERNET_DISCONNECTED during Maria scene — Vercel edge occasionally drops the connection under rapid page flips. Re-run retry would cover.
Risk / MRM / Doc-Control / Training lists	Seeder does not populate these yet. UI pages render but the lists are empty. Only affects list screens — the AI scenes (drift, signal detector, MRM populator) call APIs directly and work.